Wednesday, July 16, 2025 2:09 PM

🔌 What is a Port in Networking?

A **port** is like a **door** inside a computer or device that allows it to **communicate with other devices** on a network (like the internet).
You can think of it like this:

- The **IP address** is like a **house address** 🏠.
- A **port** is like a **specific room** or **door number** inside the house 🚪.
- Different **services** (like websites, email, or file sharing) use **different ports** to talk to each other.

🧠 Why Do We Use Ports?

To let a computer **run multiple services** at the same time — and know where to send incoming data.
For example:

- Port **80** is used for **websites (HTTP)**.
- Port **443** is for **secure websites (HTTPS)**.
- Port **25** is for **sending email (SMTP)**.
- Port **22** is for **remote login (SSH)**.

So if you open a website, your browser talks to **port 80 or 443** on the server.

🔢 Port Numbers: Quick Overview

Range	Purpose
0 – 1023	**Well-known ports** (HTTP, SSH, FTP, etc.)
1024 – 49151	**Registered ports** (custom apps)
49152 – 65535	**Dynamic/Private ports** (temporary use)

🎯 Example in Real Life

If you visit:

http://example.com
Your browser:

- Uses the internet to go to example.com (IP address)
- Connects to **port 80** by default
- Gets the website content

🛠️ How You See Ports in Use

You can use tools like:

```
netstat -tuln
```

Or scan a computer with:

```
nmap 192.168.1.1
```

✅ Summary

Term	Meaning
Port	A number that identifies a network service
IP Address	Identifies the device itself
Port + IP	Tells data **where to go**